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Mechanical power during one-lung ventilation is associated with postoperative pulmonary complications in patients undergoing lobectomy: a single-center prospective cohort study

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Abstract

Background Retrospective analyses have associated elevated mechanical power (MP) during one-lung ventilation (OLV) with reintubation rates and emergent non-invasive ventilation. Nonetheless, the study design and chosen outcomes may limit the generalizability of these findings to postoperative pulmonary complications (PPCs) in patients undergoing thoracoscopic lobectomy. The aim of this study was to evaluate whether this association remained robust under a standardized anesthesia protocol.

Method A prospective cohort study was conducted at a tertiary hospital among adults aged 50–80 years. Respiratory signals, chest computed tomography, and blood sampling were routinely measured. MP was defined as the sum of the energy transferred to overcome airway resistance and temporarily stored in the elastic lung tissue in one minute. The primary endpoint was the association of MP and PPCs. The secondary endpoints were the occurrence of impaired postoperative oxygenation, changes in lung aeration compartments, and systemic inflammatory factors during the perioperative period.

Results Patient recruitment lasted from August 1st 2022 to July 19th 2024 and included 118 participants. The incidence of PPCs was 33.9 %. MP during OLV and normalization to lung volume in non-surgical lungs, as well as the elastic and resistive components of MP, were independently associated with PPCs (OR_{adj} = 1.67 [1.01, 2.76], $p = 0.044$; 1.68 [1.04, 2.71], $p = 0.033$; 1.81 [1.16, 3.11], $p = 0.046$; and 2.61 [1.09, 6.27], $p = 0.031$, respectively). Compared with resistive components, dynamic elastic component of MP had a stronger dominance on PPCs (contribution to pseudo- R^2 0.029 vs. 0.015). The relationship between MP and PPCs was non-linear ($p = 0.047$), with a reflection point at MP of 3.7 J.min⁻¹. Compared to those without PPCs, patients with PPCs exhibited higher perioperative IL-10 levels

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and consistently poor lung aeration—both preoperatively in the non-dependent region and postoperatively in the mid-ventral region. Changes in biomarkers did not differ between patients; however, IL-8 levels showed consistent associations with PPCs during the perioperative observation period.

Conclusion Under lung-protective ventilation, MP during OLV was independently and non-linearly associated with PPCs, driven predominantly by its dynamic elastic component and accompanied by specific inflammatory and aeration changes.

Clinical trial registration Number ChiCTR2200062173 (restarted on July 27th 2022).

Key points

- Question: In the complex setting of performing thoracoscopic lobectomy in patients receiving OLV, does the association between MP and PPCs remain robust?
- Findings: A higher MP during OLV, mainly driven by ΔP , was independently and non-linearly associated with an increased risk of PPCs.
- Meaning: In adults undergoing thoracoscopic lobectomy with lung-protective ventilation, clinical measures to minimize mechanical power may reduce the risk of PPCs.

Keywords One-lung ventilation, Mechanical power, Pneumonectomy, Postoperative pulmonary complications, Lung imaging

Introduction

Worldwide, over 600,000 patients undergo lung resection annually. Notably, up to 48.8% of patients develop postoperative pulmonary complications (PPCs), constituting the leading cause of postoperative morbidity, mortality, and prolonged hospital stay [1, 2, 3]. The application of one-lung ventilation (OLV) is associated with PPCs in one-fifth of lung resections [4]. The pathophysiological mechanisms are different for ventilated and collapsed lungs. With the increasing implementation of lung-protective ventilation during OLV, volutrauma is less common [5]. Nevertheless, further efforts to mitigate atelectrauma/tidal recruitment, capillary shear stress, biotrauma, and oxidative injury remain a critical priority [6, 7].

Mechanical power (MP), the total energy delivered to the respiratory system by the ventilator per minute (J/min), is a new predictor of ventilator-induced lung injury (VILI) [8]. By integrating multiple ventilator-related factors into a single variable, the MP may provide a comprehensive framework for understanding the complex interplay of respiratory parameters during mechanical ventilation. Therein, growing evidence suggests that MP may outperform individual ventilation parameters in predicting lung injury and mortality in patients with ARDS [9–13], as well as in indicating postoperative respiratory failure and PPCs in diverse surgical populations [11, 12, 14–16], including patients undergoing thoracic surgery.

Nevertheless, the relationship between risk factors and PPCs remains complex because of methodological design and outcome selection [17]. Prior investigations have primarily utilized retrospective designs and respiratory failure leading to reintubation or emergency non-invasive ventilation. The inherent nature of retrospective study

limits control for granular confounders such as preoperative respiratory status, surgical factors (e.g., extent of resection, OLV duration), and postoperative analgesia. These unmeasured variables may bias the observed MP-PPCs relationships [18–20]. To address these limitations, we implemented a standardized anesthesia protocol minimizing variability in confounders. The primary aim of this study was to prospectively evaluate the association between intraoperative MP during OLV and PPCs in adults who underwent thoracoscopic lobectomy. We hypothesized that elevated MP during OLV is independently associated with increased PPCs risk.

Methods

Study design

We performed a prospective cohort study at the Affiliated Hospital of Southwest Medical University between August 1st 2022, and July 19th 2024, which was approved by the Ethics Committee of the affiliated hospital of Southwest Medical University, Luzhou, China (approval number: KY2022162) and the REporting of studies Conducted using STrengthening the Reporting of OBservational studies in Epidemiology (STROBE) were followed. The trial was registered with Chicttr.org.cn on July 27th 2022 (Clinical Trial number ChiCTR2200062173). Written informed consent was obtained from all participating patients. The protocol was published [21], and amendments and changes to the trial protocol are shown in *Supplement 1*.

Inclusion and exclusion criteria

Patients aged 50–80 years with an ASA of 1–2 who were scheduled for thoracoscopic lobectomy were evaluated for inclusion in this study. To ensure sufficient exposure to OLV, we excluded procedures lasting for < 1 h. Patients

were excluded if either of the following: body mass index (BMI) of $\geq 35 \text{ kg/m}^2$, thoracic surgery history, radiation therapy or chemotherapy history, critical illness, and the following lung diseases: severe pathologic lung function ($\text{FEV}_1 < 60\%$ or $\text{FVC} < 60\%$), acute lung injury, severe bronchiectasis, or emphysema. Additional exclusion criteria included COVID-19 infection history in the past half year before enrollment was added from the restart of enrollment on July 1st 2023. Patients were dropped out if surgery was converted to thoracotomy, bilobectomy, or segmental resection, or if severe hypoxemia persisted unless intermittent two-lung ventilation (TLV) was resumed.

Perioperative management of anesthesia

All patients underwent standardized anesthesia and perioperative management, including preoperative smoking cessation and respiratory exercises. The fraction of inspired oxygen (FiO_2) was set at 1.0 during pre-oxygenation and induction. General anesthesia was sequentially induced using Midazolam ($0.05 \text{ mg}\cdot\text{kg}^{-1}\cdot\text{PBW}$), Sufentanil ($0.3\text{--}0.5 \text{ mg}\cdot\text{kg}^{-1}\cdot\text{PBW}$), Propofol ($1\text{--}2 \text{ mg}\cdot\text{kg}^{-1}\cdot\text{PBW}$) and rocuronium bromide ($0.6 \text{ mg}\cdot\text{kg}^{-1}\cdot\text{PBW}$). Double-lumen tube (left 35/37Fr, COVIDIEN, Covidien Ireland Limited, IDA Business & Technology Park, Tullamore) was used to secure the airway, with the size of the catheter chosen following the principle of least resistance as the catheter lumen enters the main bronchus. A fiberoptic bronchoscope was used to position the tubes. Volume-controlled ventilation, delivered by a Mindray SV800 ventilator (Mindray Biomedical Electronics Co., Ltd., Shenzhen, China), was used for mechanical ventilation. Total intravenous anesthesia using propofol and remifentanyl was used to maintain balanced analgesia to reach the anesthetic depth within the range of 40–60 according to the bispectral index. Rocuronium was continuously infused to obtain a train of four responses between 0 and 1 using neuromuscular monitoring (M-NMT; GE Healthcare, Helsinki, Finland). Crystalloids were infused at a rate of $3\text{--}4 \text{ mg}\cdot\text{kg}^{-1}\cdot\text{h}^{-1}$.

Lung-protective ventilation strategy was employed over the operation with respiratory variables set as follows: Tidal volume (V_T) of 6–8 and $5 \text{ ml}\cdot\text{kg}^{-1}$ of predicted body weight [PBW] during TLV and OLV periods, respectively, and a constant positive end-expiratory pressure (PEEP) of $5 \text{ cmH}_2\text{O}$. Inspiratory to expiratory ratio (I: E) was 1:2. Air flow was set at $30 \text{ l}\cdot\text{min}^{-1}$ for TLV and $20 \text{ l}\cdot\text{min}^{-1}$ for OLV. The respiratory rate (RR) was adjusted to maintain end-tidal CO_2 within $42\text{--}48 \text{ cmH}_2\text{O}$, and FiO_2 was titrated to ensure peripheral oxygen saturation (SpO_2) $\geq 92\%$.

At the end of the surgery, an alveolar recruitment maneuver (ARM) was performed on both lungs using a stepwise PEEP incremental method under

pressure-controlled ventilation mode. ARM was conducted at a constant driving pressure of $10 \text{ cmH}_2\text{O}$ from a PEEP of $5 \text{ cmH}_2\text{O}$, increasing by $5 \text{ cmH}_2\text{O}$ every 15 s until a peak inspiratory pressure of $30 \text{ cmH}_2\text{O}$ was reached (detailed in Supp. Figure 2). Subsequently, the TLV was resumed, and V_T was targeted at $6 \text{ mL}\cdot\text{kg}^{-1}\cdot\text{PBW}^{-1}$ due to the reduction of lung capacity postoperatively.

During recovery from general anesthesia, the ventilation mode was switched to pressure-synchronized intermittent mandatory ventilation mode until extubation was achieved. RR was dynamically adjusted using an inspiratory trigger window of 20%. Multimodal analgesia was administered to ensure adequate postoperative pain control. Further details can be found elsewhere [21].

Rescue strategies

If hypoxemia ($\text{SpO}_2 < 92\%$ for 2 min) [22] occurred after establishing the artificial airway but before the initiation of OLV, inspiratory holding was applied at $30 \text{ cmH}_2\text{O}$ to both lungs for 15–20 s to treat atelectasis generated during mask ventilation and tracheal intubation.

If hypoxemia occurred during OLV during surgery, interventions were introduced in the following order: increased FiO_2 , continuous positive airway pressure to the surgical lung (no more than $5 \text{ cmH}_2\text{O}$), ligation of one pulmonary artery, and intermittent TLV [4].

If the peak pressure exceeded $30 \text{ cmH}_2\text{O}$, the inspiratory flow was reduced after ruling out double-lumen dislocation. If the driving pressure (ΔP) during OLV exceeded $15 \text{ cmH}_2\text{O}$, ARM with stepwise incremental PEEP was performed to improve lung compliance. Bronchodilators were administered if airway pressure remained persistently high.

Measurements

Respiratory signals, including peak airway pressure (P_{peak}), plateau pressure (P_{plat}), mean airway pressure (P_{mean}), PEEP, VT, RR, and flow, were recorded from the screen of the ventilator at the following time points: initiating mechanical ventilation ($\text{TLV}_{\text{pre-op}}$), the start of OLV and surgery in the lateral decubitus position ($\text{OLV}_{\text{start}}$), one hour after ($\text{OLV}_{1\text{h}}$), and resumption of TLV in the supine position after surgery ($\text{TLV}_{\text{post-op}}$). The time course is detailed in the Supplementary Material section.

ΔP was calculated as P_{plat} minus PEEP. Blood gas analysis was routinely performed upon admission to the hospital, and measurements were taken using a commercially available device at $\text{OLV}_{1\text{h}}$ during the operation and 30 min after admission to the postoperative anesthesia care unit (ABL80, Radiometer, Denmark).

MP calculation

MP was calculated using a formula similar to the one proposed by Gattinoni et al. [8], but explicitly excluding the static component of power related to PEEP (Eq. 1), as it represents energy stored to maintain functional residual capacity rather than directly contributing to cyclic tidal inflation [10,23]. This methodological approach was based on the pure physical basics of a uniaxial elastic spring, as described elsewhere [10]. The implications of this choice, including its effect on absolute MP values and comparability with other studies, are addressed in the Discussion.

$$MP = 0.098 \bullet RR \bullet V_T^2 \bullet \left[\frac{1}{2} \bullet E_{rs} + RR \bullet \frac{(1 + I : E)}{(60 \bullet I : E)} \bullet R_{aw} \right] \quad (1)$$

$$MP_e = 0.098 \bullet RR \bullet V_T^2 \bullet \frac{1}{2} \bullet E_{rs} \quad (2)$$

$$MP_r = 0.098 \bullet RR \bullet V_T^2 \bullet \frac{(1 + I : E)}{(60 \bullet I : E)} \bullet R_{aw} \quad (3)$$

where I: E stands for inspiratory to expiratory ratio, and E_{rs} and R_{aw} stand for respiratory system elastance and airway resistance, respectively, and are calculated as follows:

$$E_{rs} = \frac{P_{plat} - PEEP}{V_T} \quad (4)$$

$$R_{aw} = \frac{(P_{peak} - P_{plat}) \bullet 60}{Flow} \quad (5)$$

MP measured at TLV_{pre-op} and OLV_{1h} were further normalized to the preoperative lung volume of both lungs (norMP_{TLV}) and non-surgical lung (norMP_{OLV}), respectively.

$$\text{norMP}_{TLV} = \frac{MP_{TLV_{pre-op}}}{\text{Lung volume}_{two\ lungs}} \quad (6)$$

$$\text{norMP}_{OLV} = \frac{MP_{OLV_{1h}}}{\text{Lung volume}_{non-surgical\ lung}} \quad (7)$$

Chest radiography

Pre- and postoperative chest computed tomography (CT) scans were examined with end-inspiratory breath-holds of approximately 10 s in a blinded manner by an independent specialist in radiology, who was not involved in the study. Pathological CT was defined as the presence of at least one of the following: atelectasis, pleural effusion, or other radiological chest alterations.

Assessment of lung aeration

CT images were reconstructed using a B60F convolution kernel, producing a 512 × 512 matrix with 0.5 mm × 0.5 mm × 1 mm voxel size. Lung segmentation was performed using a deep convolutional neural network algorithm (Shenzhen Yitu Intelligent Technology Co., China), followed by manual refinement. The lung volume was calculated by multiplying the pixel count by the pixel size and slice thickness. Aeration compartments were classified into four categories based on the Hounsfield units and were expressed as a percentage of the total lung mass. Four equidistant horizontal regions of interest along the ventro-dorsal axis were defined as the ventral, mid-ventral, mid-dorsal, and dorsal regions (Supp. Figure 3).

Blood sampling and measurements

Originally, venous blood samples were obtained before induction of anesthesia (baseline), immediately after extubation (POD₀) and on the 1st (POD₁) and 2nd postoperative day (POD₂). The samples were centrifuged at 1500–2000 rpm for 15 min, and the supernatant was stored at -80 °C. After the plasma samples had been sent to a biotechnology company (Sainty Biotech Co., Ltd., Chengdu, China), and adequate sample quality was confirmed, interleukin (IL)-6, IL-8, IL-10, and tumor necrosis factor (TNF)-α were measured by ELISA (Elabscience Biotech Co., Ltd., Wuhan, China).

Postoperative observations

PPCs were defined as the development of one of the following findings whatever occurred first using Standardized Endpoints for Perioperative Medicine (StEP) [24]: Atelectasis, Pneumonia, Acute Respiratory Distress Syndrome (ARDS, using Berlin definition), and pulmonary aspiration. Deviating from the initially registered protocol that defined PPCs using the Melbourne Group scale (MGS), this approach aimed to incorporate low-severity pulmonary events and isolated complications into the overall clinical outcome assessment, particularly in the context of minimally invasive thoracic surgery. Details about the reassessment are discussed in the Supplemental materials.

Impaired postoperative oxygenation was defined as at least one of the following: (1) SpO₂ < 90%; or (2) imputed PaO₂/FiO₂ < 300 mmHg [either (1) or (2) occurring at least twice within 24 h]; or (3) high nasal cannula oxygen flow to deliver FiO₂ ≥ 0.5; or (4) reintubation and invasive mechanical ventilation.

Primary and secondary endpoints

The primary endpoint was the association of MP and PPCs. The secondary endpoints were the occurrence of impaired postoperative oxygenation, changes in lung

aeration compartments, and systemic inflammatory factors during the perioperative period.

Statistical analysis

The sample size was calculated a priori by using a two-tailed independent t-test (PASS 2022 software). At the time of study planning, no published data was available regarding the expected effect size or variability of the association between MP (as a continuous variable) and PPCs in patients undergoing thoracoscopic lobectomy. Therefore, the calculation was pragmatically based on differences in MP itself, using pilot data that indicated a risk ratio of 3.375 between high and low MP groups. Assuming a dropout rate of 10%, a sample size of 120 patients would provide an appropriate power ($1-\beta=0.9$) to identify a significant difference [21]. However, we planned a *post hoc* power analysis based on the final cohort's observed parameters.

Categorical variables are presented as numbers (percentages), while continuous variables are expressed as mean $\pm$ SD or median [IQR] for parametric and non-parametric data. Group comparisons were performed using Chi-square/Fisher's exact tests and t-tests/Mann-Whitney U tests for categorical and continuous variables, respectively. A general linear model was employed for repetitive data. Continuous confounders were categorized into clinically relevant categories. Associations were determined using multivariate logistic regression, expressed by OR [95% CI]. Given the known and severe multicollinearity between MP and its components (ΔP , MP_e , MP_r), as well as the normalized MP, we constructed separate multivariable logistic regression models for each ventilatory parameter to avoid estimation bias. Each separate model was adjusted for a consistent set of statistically significant confounders in the univariate logistic regression. A generalized additive model to identify the nonlinear relationship [25,26] and a two-piecewise linear regression model was used to calculate the threshold effect of MP during OLV on PPCs in terms of the smoothing plot. When MP during OLV and PPCs was evident in the smoothed curve, the recursive method automatically calculated the threshold, and the maximum model likelihood was used [25]. We performed tests for linear trends by entering the median value of each category of MP as a continuous variable in the models.

Dominance analysis was employed to evaluate and compare the relative importance of MP in explaining the PPCs variance. The incremental R^2 contribution of predictors to the full model was measured by interactional dominance [27]. Due to the prospective design, all ventilatory parameters were captured electronically in real-time from ventilators, ensuring no missing data for these variables. Thus, no imputation was needed and analyses were performed on the complete dataset using R

Foundation (<http://www.R-project.org>) and the Free Statistics Analysis Platform (version 1.8, China). Statistical significance was set at a two-tailed p -value < 0.05 .

Results

Study population and general features

A total of 120 participants were enrolled in this study. Two patients were excluded because of the implausibility of airway pressure recorded at TLV_{pre-op} and OLV_{1h} ; thus, 118 patients were analyzed. The average durations of OLV and general anesthesia were 104 and 170 min, respectively. Data on the cohort characteristics and intra-operative respiratory function are shown in (Table 1; Fig. 1), respectively. The incidence of PPCs was 33.9% and pneumonia was the predominant complication (82.5%). Atelectasis was found in 27.5% of patients who developed PPCs. None of the patients in this cohort experienced ARDS or pulmonary aspiration. The occurrence of PPCs has extended hospitalization. In patients with PPCs, hypertension and right-sided lobectomy were more common and more frequently experienced moderate pain on POD_2 . In addition, propofol consumption during anesthesia was greater, higher FiO_2 was required during OLV, and $PA-aDO_2$ measured at OLV_{1h} was greater (Table 1).

Correlation between MP during OLV and PPCs

Higher BMI, comorbid hypertension, right surgical site, greater propofol consumption, respiratory parameters (P_{peak} , P_{plat} , ΔP , resistance, MP and $norMP_{OLV}$, MP_e , and MP_r), and FiO_2 at OLV_{1h} were associated with an increased risk of PPCs (Supp. Table 1). However, after adjustment, MP and $norMP_{OLV}$, MP_e , and MP_r , but not airway pressure, were significantly associated with PPCs (Table 2). As determined by its contribution to PPCs according to dominance analysis, MP_e during OLV was more dominant than MP_r (contribution to pseudo- R^2 0.029 vs. 0.015). In patients developed PPCs, $norMP_{OLV}$ (2.2 ± 1.1 vs. 1.8 ± 0.7 , $p=0.004$) but not $norMP_{TLV}$ (1.0 ± 0.4 vs. 0.9 ± 0.3 , $p=0.133$) was higher than patients without.

Other endpoints

The occurrence of impaired postoperative oxygenation

More than half of patients experienced impaired oxygenation (50.9%), and the occurrence was twice as often in patients with PPCs as without (75% vs. 38.5%, $p < 0.001$).

Changes in lung aeration compartments according to the occurrence of PPCs

All patients experienced significant changes in lung aeration after surgery (Supp. Figure 4). Compared with patients without PPCs, normally aerated compartments in non-dependent lungs were lower (ventral region: 75.36 [63.73,78.9] vs. 80.55 [71.9,82.68] %, $p < 0.001$).

Table 1 Description of characteristics in patients with vs. without PPCs

General characteristics	Non-PPCs(n =78)	PPCs(n =40)	P-value
Perioperative factors			
Male gender	28 (35.9)	20 (50)	0.140
Age (years)	60.1 ± 7.0	62.0 ± 7.8	0.181
Height (cm)	158.6 ± 7.7	160.3 ± 6.4	0.221
Weight (kg)	60.2 ± 11.0	64.2 ± 9.1	0.047
BMI (kg.m ⁻²)	23.8 ± 3.4	24.9 ± 2.8	0.076
FEV ₁ (%)	78.7 ± 7.0	79.1 ± 6.0	0.739
FEV ₁ / predicted FEV ₁ (%)	95.7 ± 10.6	98.5 ± 8.6	0.143
ARISCAT score	34.4 ± 9.5	36.5 ± 9.5	0.264
ARISCAT classification			0.760
Intermediate risk	70 (89.7)	35 (87.5)	
High risk	8 (10.3)	5 (12.5)	
Perioperative risk factors for PPCs			
Preoperative SpO ₂ (%)	96.8 ± 1.1	96.0 ± 1.7	0.002
Preoperative Hemoglobin (g.dl ⁻¹)	127.8 ± 15.2	132.8 ± 15.5	0.094
Respiratory infection within the last month	18 (23.1)	12 (30)	0.414
Active smoker	26 (33.3)	16 (40)	0.474
ASA physical status classification			0.970
1 (best health)	6 (7.7)	3 (7.5)	
2	72(92.3)	37(92.5)	3
Hypertension	24 (30.8)	20 (50)	0.041
Coronary artery disease	3 (3.8)	3 (7.5)	0.406
Diabetes	16 (20.5)	6 (15)	0.467
Surgical site (right)	48 (61.5)	33 (82.5)	0.020
Surgical scope			0.376
Upper lobectomy	42 (53.8)	26 (65)	
Middle lobectomy	11 (14.1)	6 (15)	
Lower lobectomy	25 (32.1)	8 (20)	
Duration of anesthesia (min)	167.1 ± 35.0	176.6 ± 39.5	0.181
Duration of OLV (min)	102.4 ± 32.1	107.0 ± 35.2	0.482
Vasopressor use	38 (48.7)	18 (45)	0.702
Rescue strategy			
RMs before surgery	1 (1.3)	1 (2.5)	1
Increasing FiO ₂	2(2.6)	3(7.5)	0.335
CPAP to surgical lung	0(0)	1(2.5)	0.339
Ligation of one pulmonary artery	0(0)	1(2.5)	0.339
Preoperative lung volume (ml)			
Both lungs	4372.8 ± 1291.0	4457.9 ± 1196.3	0.729
Ventilated lung	2165.6 ± 615.5	2139.6 ± 607.0	0.827
Anesthetic usage and Fluid balance			
Remifentanyl (μg.kg ⁻¹)	13.4 (11.2, 17.0)	14.1 (11.1, 18.7)	0.560
Propofol (mg.kg ⁻¹)	9.7 (7.6, 12.6)	11.9 (9.5, 14.6)	0.032
Muscle relaxant (mg.kg ⁻¹)	0.1 (0.0, 0.2)	0.1 (0.0, 0.2)	0.555
Norepinephrine (μg.kg ⁻¹)	0 (0. 3.7)	0 (0. 3.5)	0.945
Crystalloid (ml.kg ⁻¹)	5 (4.0, 6.2)	5.3 (4.3, 6.0)	0.560
Blood loss (ml.kg ⁻¹)	0.4 (0.3, 0.6)	0.3 (0.3, 0.4)	0.456
Gas exchange during OLV			
FiO ₂ (%)	50.1 ± 13.2	57.1 ± 18.0	0.018
PA-aDO ₂ (mmHg)	197.0 ± 92	256.2 ± 123.4	0.026
In-hospital length (day)	4.0 [4.0, 6.0]	4.0 [4.0, 5.0]	0.022

ARISCAT Assess Respiratory Risk in Surgical Patients in Catalonia, BMI/Body mass index, FEV₁ Forced expiratory volume in one second, OLV One-lung ventilation, PA-aDO₂ Partial pressure of alveolar gas-arterial blood oxygen, RM Recruitment maneuver

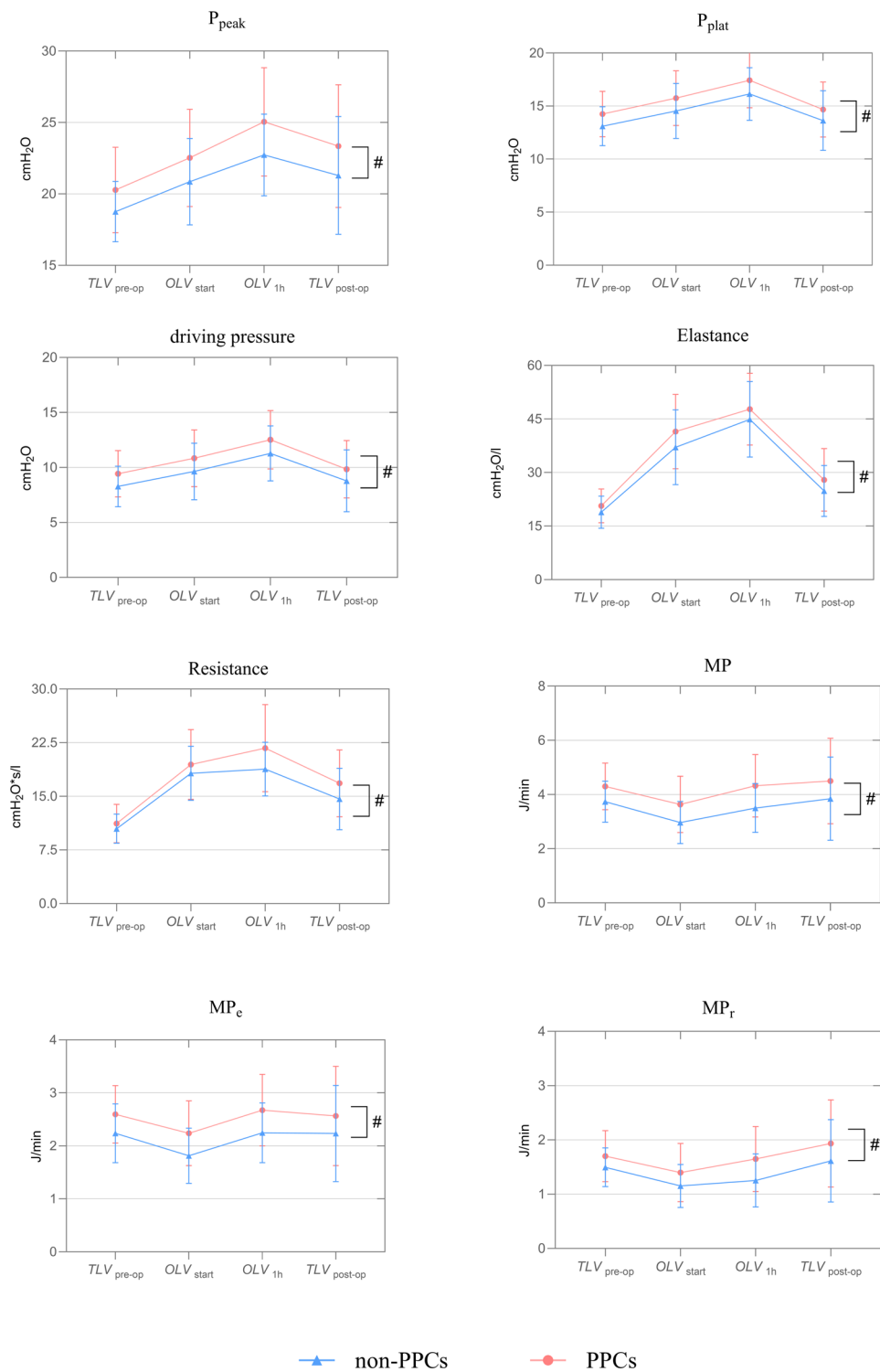


Fig. 1 Time course of respiratory variables throughout the intervention time. MP_e Elastic component of MP, MP_r Resistive component of MP, TLV_{pre-op} Two-lung ventilation in the supine position before surgery, OLV_{start} the start of one-lung ventilation and surgery in the lateral decubitus position, OLV_{1h} One hour after the initiation of one-lung ventilation, $TLV_{post-op}$ Two-lung ventilation after surgery in the supine position. #represents a significant difference between patients with and without PPCs (GLM)

Table 2 Multivariable logistic regression analysis of MP and PPCs

Model	Variables	OR [95%CI]	P-value
Model 1	ΔP^b	1.01 [0.83, 1.23]	0.928
Model 2	MP ^b	1.67 [1.01, 2.76]	0.044
Model 3	MP _e ^b	1.81 [1.16, 3.11]	0.046
Model 4	MP _r ^b	2.61 [1.09, 6.27]	0.031
Model 5	norMP _{OLV}	1.68 [1.04, 2.71]	0.033
Model 6	norMP _{TLV}	2.78 [0.83, 9.29]	0.098

Models were adjusted for variables with statistical significance in univariate logistic regression (including BMI, hypertension, surgical site, propofol, and FiO₂ at OLV_{1h}). ΔP Driving pressure, MP Mechanical power, MP_e Elastic component of MP, MP_r Resistive component of MP, norMP_{OLV} and norMP_{TLV} are the normalization of MP measured at TLV_{pre-op} and OLV_{1h} to the preoperative lung volume of both lungs and ventilated lungs, respectively. ^b, measurement taken one hour after the start of OLV

$p = 0.013$; mid-ventral region: 82.12 [73.54,85.18] vs. 86.5 [81.56,88.36] %, $p = 0.032$) and atelectasis in the mid-ventral region was higher (0.78 [0.34,1.06] vs. 0.31 [0.21,0.65] %, $p = 0.019$) in patients with PPCs before surgery, whereas the normally aerated compartment in the mid-ventral region remained lower after surgery (81.72 [74.7, 85.69] vs.85.2 [80.73, 88.54], $p = 0.020$, Fig. 2).

Changes in levels of biomarkers according to the occurrence of PPCs

Biomarker levels in patients with and without PPCs are presented in Table 3. After surgery, the concentrations of IL-6 and IL-8 measured on POD₂, and IL-10 measured on POD₀₋₂ were higher in patients with PPCs. However, the relative changes in biomarker levels did not show a statistically significant difference between them (Supp. Figure 5).

Explorative analysis

Non-linear correlation

We observed a nonlinear dose-response relationship between PPCs and MP during OLV (Supp. Figure 6, p for non-linearity = 0.047), where the inflection point of PPCs risk was an MP of 3.7 J.min⁻¹. Above this threshold, PPCs increased rapidly and remained significantly associated with MP, whereas below the threshold, the dose-response curve lost statistical significance (Table 4).

Correlation between biomarkers with PPCs

Biomarkers measured either pre-operatively or immediately after extubation were not associated with PPCs in the crude model, whereas IL-8 levels exhibited robust associations across multiple time points in the adjusted models. TNF- α was not significantly associated with PPCs in either model (Supp. Table 2).

Discussion

Among adult patients undergoing thoracoscopic lobectomy for lung cancer, we found that: (1) A higher MP during OLV, mainly driven by ΔP , was independently

associated with an increased risk of PPCs. (2) This association remained significant when MP levels exceeded a threshold of 3.7 J.min⁻¹. (3) Compared to those without PPCs, patients with PPCs exhibited fewer normally aerated compartments and more severe atelectasis in non-dependent lung regions preoperatively, and maintained fewer normally aerated compartments in the mid-ventral region postoperatively. (4) Changes in biomarkers did not differ between patients; however, perioperative elevations in IL-8 were independently associated with PPCs risks.

Studies have reported that higher intraoperative MP is associated with increased PPCs in non-thoracic surgery and severe postoperative events, i.e., reintubation and non-invasive ventilation, in thoracic surgery [11,12,14,15]. Our findings contribute to the growing body of evidence that MP is a pivotal integrative variable in VILI pathogenesis, capable of identifying lower-severity pulmonary events and isolated complications during the relatively short period of OLV. The association of norMP_{OLV} with PPCs further enhanced the clinical merit of MP.

Consistent with previous studies, a higher MP, mainly driven by ΔP , was independently associated with an increased risk of PPCs, indicating that the dynamic elastic component was a more informative component of PPCs [12]. This is physiologically plausible, as during OLV, the dependent lung (non-surgical side) is predisposed to atelectasis and elevated pleural pressure, thereby reducing its functional volume and making it more susceptible to atelectrauma, while high dynamic elastic power indicates more severe effects of these destructive shear forces. However, ΔP itself failed to show a significant association with PPCs despite it was higher in patients with PPCs over the operation period. This finding aligns with existing evidence in cardiothoracic surgery [28] and with evidence that attempts in thoracic surgery to reduce ΔP during OLV to decrease PPCs have been ineffective [29]. This suggests that atelectrauma exhibits energy-intensive properties and is directly amplified by a high cycling frequency and high stress.

The non-linear relationship we identified is consistent with previously reported U-shaped dose-response function between MP and postoperative respiratory failure. We identified a significant independent association between higher MP and PPCs only when MP levels exceeded a threshold of 3.7 J.min⁻¹. This observation partly explains the inconsistent correlations noted in both infant and adult populations undergoing thoracic surgery, with the association being absent in infants (the median MP of 2.17 J.min⁻¹) but present in adults (the median MP of 8.5 J.min⁻¹) [11, 30]. This inconsistency suggests that the indicative value of MP may be constrained by its absolute magnitude and exhibits limited performance in populations with low values.

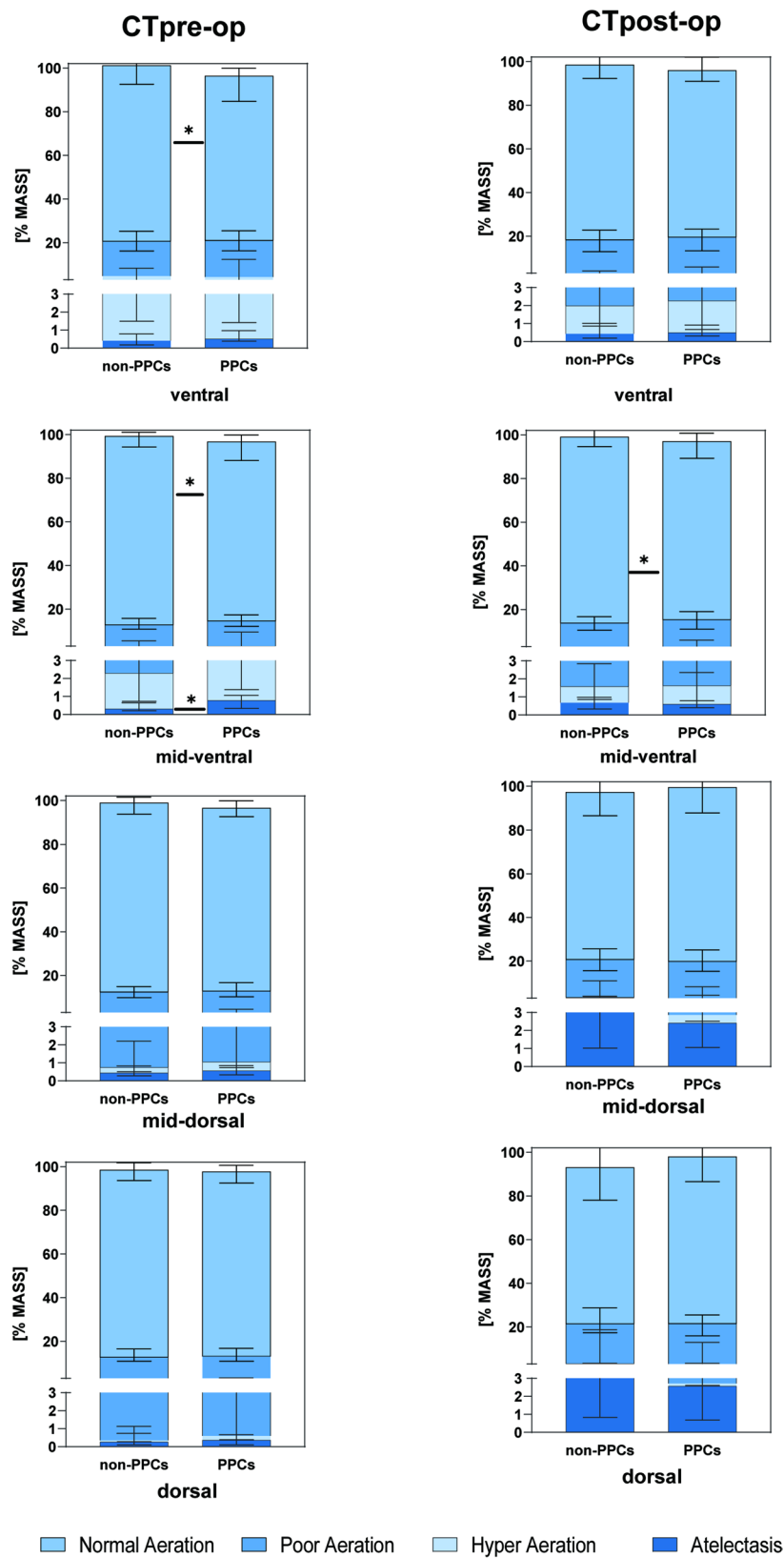


Fig. 2 (See legend on next page.)

(See figure on previous page.)

Fig. 2 Comparison of aeration compartments in the non-surgical lung between patients with and without PPCs on four iso-gravitational planes. Lung aeration compartments were expressed as a percentage of the total lung mass [% mass]. Histograms are reported as media with an IQR. The hyper-aerated compartment is plotted in white blue, the normally aerated compartment in light blue, the poorly aerated compartment in blue, and the atelectasis compartment in dark blue. The Mann-Whitney U test was employed for the comparison between two independent groups. Statistically significant differences in aeration compartments between two independent groups were labelled by *

A fundamental consideration in interpreting our findings is our methodological choice to calculate MP without the static PEEP-related component. While this allows us to isolate the dynamic energy imparted by tidal ventilation, it has direct consequences. First, since PEEP power accounts for approximately half of the total power [31], the absolute values of MP reported in this paper are consequently lower than the formula-derived values that include PEEP ($\approx 6\text{--}12 \text{ J}\cdot\text{min}^{-1}$). Therefore, direct numerical comparison of our MP values or the identified $3.7 \text{ J}\cdot\text{min}^{-1}$ threshold with those from studies employing different formulas is invalid. Second, the clinical relevance of the excluded PEEP power remains debated. While it may represent ‘useful’ energy to maintain alveolar patency, its inclusion may alter the predictive power of MP for certain outcomes. In both a severe ARDS pig model (induced by saline lavages and injurious ventilation) [32] and a moderate ARDS pig model (induced by saline lavages alone) [10], lung injury was measured by the uptake rate of [18] F-FDG via positron-emission tomography. In each model, MP containing driving power and resistive power (Eq. 1) instead of total power (containing PEEP power) showed positive association with neutrophilic inflammation. Nevertheless, when assessing lung injury through pulmonary histology, inflammatory biomarkers, alveolar tension, and damage to epithelial or endothelial cells, elastic power (defined as the sum of PEEP power and drive power) provides a more quantitative reflection of injury severity than either total MP or drive power alone [33]. Future research validating MP thresholds for clinical use must account for and standardize this methodological variation.

We also found significant associations between PPCs and higher BMI, hypertension, right-sided lobectomy, increased propofol dosage, and higher FiO_2 during the OLV. Overweight/obese patients are prone to decreased chest wall compliance, diminished lung capacity, and elevated oxygen demand, leading to the development of PPCs [34,35], which may be one of the reasons for the doubling of incidence of pneumonia in patients with $\text{BMI} > 25$ in our study (40.4 vs. 19.7%, $p = 0.014$). Due to anatomical factors, including greater functional parenchyma loss, compromised lymphatic drainage (particularly in upper lobe resection), and technical complexity during right upper/middle lobe resections, right-sided lobectomy showed a 1.95-fold increased PPCs risk compared to left-sided procedures. This finding aligns with STS database showing the highest rate for right upper

and lower lobes and the lowest rate for right middle lobe (8.94 vs. 9.32 vs. 5.36%) [36]. The harmful effect of high intra-operative FiO_2 has recently been quantified as chemical power and shown to be independently associated with PPCs. Each $1 \text{ J}\cdot\text{min}^{-1}$ increase in chemical power was linked to an 8% rise in PPCs, while each $1 \text{ J}\cdot\text{min}^{-1}$ increase in MP was related a rise of 5% [37], highlighting that both forms of energy contribute independently to pulmonary harm.

Our study exhibited the dual role of inflammatory mediators in PPCs pathogenesis. The persistent elevation of proinflammatory biomarkers represents sustained neutrophilic activation and reactive oxygen species release, which directly cause endothelial and epithelial injury, increase vascular permeability, and contribute to the breakdown of the alveolar-capillary barrier [38]. Conversely, the persistent elevation of IL-10 suggests a protective mechanism aimed at counterbalancing excessive pro-inflammatory activity. However, IL-10 measured over POD_{0-2} were significantly higher in patients with PPCs compared to those without. This suggests that a marked and sustained anti-inflammatory response following major procedures such as thoroscopic lobectomy may indicate a transition toward a state of compensatory anti-inflammatory response syndrome [39]. Notably, the consistent perioperative IL-8 association underscores its potential as a PPCs predictor, reflecting pre-existing pro-inflammatory predisposition amplified by OLV.

Limitations

This cohort study had some limitations. First, this study was monocentric with a relatively small sample size. This may limit the generalizability of our results. Second, the shift of PPCs definition from the MGS to StEP criteria is likely to result in a higher incidence of PPCs due to the inherent high sensitivity. Therefore, our incidence rates should not be directly compared with those from studies that utilized the MGS without appropriate consideration of this methodological difference. Third, determining the sample size based on differences in MP rather than on the primary endpoint—the association between MP and PPCs—is a significant methodological limitation. This decision was made due to the absence of prior effect-size estimates for the MP–PPC relationship in this specific setting at the time of study inception. Consequently, the present study should be regarded as exploratory and hypothesis-generating rather than confirmatory. The effect estimates, and particularly the identified threshold

Table 3 Plasma levels of biomarkers at time points by occurrence of PPCs

	Non-PPCs (n = 78)	PPCs (n = 40)	P-value
IL-6 (pg ml ⁻¹)			
baseline	0.48 (0.24, 0.90)	0.65 (0.33, 1.67)	0.059
POD ₀	3.72 (1.46, 6.92)	4.41 (2.86, 11.03)	0.063
POD ₁	7.53 (3.11, 13.90)	12.02 (6.78, 22.63)	0.053
POD ₂	4.44 (2.15, 12.22)	13.07 (5.76, 19.40)	0.001
IL-8 (pg ml ⁻¹)			
baseline	3.81 (1.71, 6.80)	5.38 (2.99, 8.83)	0.075
POD ₀	4.88 (2.29, 8.03)	6.80 (3.33, 8.06)	0.194
POD ₁	5.34 (2.88, 9.35)	6.80 (3.70, 11.61)	0.137
POD ₂	5.18 (2.86, 8.76)	7.32 (4.10, 12.50)	0.036
IL-10 (pg ml ⁻¹)			
baseline	0.44 (0.24, 0.65)	0.54 (0.31, 0.92)	0.182
POD ₀	0.73 (0.39, 1.09)	1.00 (0.61, 1.52)	0.012
POD ₁	1.08 (0.65, 1.92)	1.70 (0.78, 2.85)	0.029
POD ₂	0.78 (0.48, 1.48)	1.29 (0.80, 2.07)	0.005
TNF-α (pg ml ⁻¹)			
baseline	11.13 (6.60, 32.55)	14.60 (7.77, 30.86)	0.710
POD ₀	10.29 (6.85, 27.55)	11.52 (3.94, 22.30)	0.392
POD ₁	15.86 (8.43, 33.65)	17.43 (7.30, 36.71)	1.000
POD ₂	15.40 (7.51, 35.60)	20.80 (11.26, 37.82)	0.337

Baseline, measured pre-operatively, POD₀, after extubation, POD₁ the 1st postoperative day, POD₂ the 2nd postoperative day

Table 4 Threshold effect analysis of MP during OLV on PPCs

Threshold of MP	Crude model		Adjusted model	
	OR [95%CI]	p-value	OR [95% CI]	p-value
< 3.7 J/min	10.46 [1.72, 63.48]	0.011	13.03 [0.93, 182.32]	0.057
≥ 3.7 J/min	3.52 [1.4, 8.69]	0.006	2.91 [1.08, 7.86]	0.035

Adjustment factors included BMI, hypertension, surgical site, propofol, FIO₂ during OLV. MP Mechanical power

of 3.7 J·min⁻¹, require validation in independent cohorts. Nonetheless, we performed a *post hoc* power calculation using the observed effect size (PPC incidence 0.339, OR 1.67 per 1 J·min⁻¹ increase, SD 1.1 J·min⁻¹, α = 0.05), which confirmed that 110 patients were required to achieve 80% power and our final sample of 118 patients thus provided sufficient statistical power to detect the observed effect. These findings provide the necessary preliminary data to inform a rigorous, endpoint-driven sample size calculation for a future multicenter confirmatory trial. Fourth, a stepwise incremental PEEP maneuver utilizing a P_{peak} of 30 cmH₂O was employed to facilitate the recruitment of surgically collapsed lung tissue. The adequacy of this pressure level for complete lung recruitment remains debatable [40,41], however, thoroscopic observations ensured effective recruitment.

Conclusions

In adults undergoing thoroscopic lobectomy with lung-protective ventilation, PPCs were nonlinearly associated with MP during OLV and independently with perioperative elevations in IL-8. Patients with persistently poor lung aeration were prone to developing PPCs. An integrated strategy targeting intraoperative ventilation and preoperative risk stratification is warranted.

Abbreviations

ARDS	Acute respiratory distress syndrome
ARM	Alveolar recruitment manoeuvre
COPD	Chronic obstructive pulmonary disease
CT	Computed tomography
E _{rs}	Elastance of respiratory system
F _I O ₂	Inspiratory oxygen fraction
HR	Heart rate
I: E	Inspiratory to expiratory time ratio
MAP	Mean arterial blood pressure
MGs	Melbourne Group scale
MP	Mechanical power
PBW	Predicted bodyweight
PaO ₂	Arterial partial pressure of oxygen
PaO ₂ /F _I O ₂	Horovitz-index
PEEP	Positive end-expiratory pressure
P _{peak}	Peak airway pressure
P _{plat}	Plateau pressure
R _{aw}	Resistance of airway
ROIs	Regions of interest
RR	Respiratory rate
VATS	Video-assisted thoracic surgery
VCV	Volume controlled ventilation
VILI	Ventilator-induced lung injury
V _T	Tidal volume
ΔP	Driving pressure

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12871-026-03776-1>.

Supplementary Material 1.

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Authors' contributions

YYZ, LL, YPB, XYL, and XBW designed the study. LL and YPB executed the protocol and acquired the clinical data. YYZ, YMZ, YPB, and XW performed the study. YYZ, XW, ZYZ, and MS were involved in the data analysis and/or interpretation. YYZ, XW, and ZPD performed the image data preprocessing and analysed the CT image acquisition. YYZ drafted the manuscript. LL, MS and XBW reviewed and critically revised the manuscript. All authors have read and approved the submitted manuscript and agreed to be accountable for the content of the work and its publication.

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had no involvement in the study design, data collection, data analysis, data interpretation, or writing of the manuscript.

Data availability

The datasets generated and analyzed in this study will be made available by the authors upon personal request from the corresponding author. The supplemental digital file describes the detailed methodology, as well as the amendments and changes to the trial protocol and registration.

Declarations

Ethics approval and consent to participate

The ethics committee of the Affiliated Hospital of Southwest Medical University, China approved this study (reference number: KY2022162). All participants provided informed consent before participation.

Consent for publication

Not applicable.

Competing interests

The authors declare no conflict of interests.

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